

Computer Organization and Assembly Language

Class Test: 2
Time: 30 minutes
ROLL NO :

BSE F23 MOR
Instructor: Abdul Khaliq

Question 1:

[10]

Consider the Data area of a Program given below (where the address of first Variable i.e V1V is 0013) How this data will be stored in the memory (fill the given area with suitable contents in hexadecimal form at given addresses)

V1V DB 0ABh, 0CDh, 0EFh, 40h, 50h, 60h
V2V DB -32
V3V DB 123D, 1011B
V4V DW 1234h, 0ABCDh
V5V DD 45768923h, 1234ABCDh
V6V DW 10 Dup(?)
V7V DB "AAAAA"
V8V DW V2V

(Note : ASCII code of character A is 41h)

0010	0011	0012	0013	0014	0015	0016	0017	0018	0019	001A	001B	001C	001D	001E	001F
			AB	CD	EF	40	50	60	E0	7B	0B	34	12	CD	AB
0020	0021	0022	0023	0024	0025	0026	0027	0028	0029	002A	002B	002C	002D	002E	002F
23	89	76	45	CD	AB	34	12	00	00	00	00	00	00	00	00
0030	0031	0032	0033	0034	0035	0036	0037	0038	0039	003A	003B	003C	003D	003E	003F
00	00	00	00	00	00	00	00	00	00	00	00	41	41	41	41
0040	0041	0042		...	...	...	...	...	...	...	...	...	...	...	004F
41	19	00													

Computer Organization and Assembly Language

Question 2:

[10]

Consider the same Data area of the Program (discussed earlier), Give the Values of the mentioned operand, after the execution of the given instructions

	INSTRUCTION	OUTPUT
1	MOV AL,0AH	0A
2	MOV BL,[0017]	50
3	MOV CL , BYTE PTR V4V	34
4	MOV AH, Type V4V	02
5	MOV BH, LENGTH V6V	10
6	MOV CH, SIZE V6V	20
7	MOV AX,V8V	0019
8	MOV BX,V4V +2	ABCD
9	MOV CX,WORD PTR V5V+2	4576
10	MOV DX,OFFSET V7V	003C

Question 3:

[5]

As each of the instruction in the following program is executed, fill in the hexadecimal value of the operand listed on the right hand side:

```

.data
Var1 dw 0cdefh
Var2 dw 3456h

.code
:
    Mov ax,var1

    Xchg var2,ax    ; AX=3456

    Dec ax

    Sub var2,2      ;VAR2= cded

    Mov bx,var2

    Add ah,bl       ;AX=2155
:
:

```
